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WEEK-6

One-Dimensional Arrays

Ex.No. 34

Date 15.11.24

Check pair with difference k**Problem Statement:**

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[i] - A[j] = k$, $i \neq j$.

Input Format

1. First line is number of test cases T. Following T lines contain:
2. N, followed by N integers of the array
3. The non-negative integer k

Output format

Print 1 if such a pair exists and 0 if it doesn't.

Sample Input:

```
1
3 1 3 5
4
```

Sample Output:

```
1
```

Program:

```
1  #include<stdio.h>
2  int main()
3  {
4      int t;
5      scanf("%d",&t);
6      while(t-->0)
7      {
8          int n;
9          scanf("%d",&n);
10         int a[n];
11         for(int i=0;i<n;i++)
12         {
13             scanf("%d",&a[i]);
14         }
15         int k;
16         scanf("%d",&k);
17         int flag=0;
18         for(int i=0;i<n;i++)
19         {
20             for(int j=i+1;j<n;j++)
21             {
22                 if(a[i]-a[j]==k || a[j]-a[i]==k)
23                 {
24                     flag=1;
25                     break;
26                 }
27             }
28             if(flag)break;
29         }
```

```
29     }  
30     printf("%d\n",flag);  
31 }  
32 }
```

	Input	Expected	Got	
✓	1 3 1 3 5 4	1	1	✓
✓	1 3 1 3 5 99	0	0	✓

Passed all tests! ✓

Ex.No. 35

Date 15.11.24

Chocolates

Problem Statement:

Sam loves chocolates and starts buying them on the 1st day of the year. Each day of the year, x , is numbered from 1 to Y . On days when x is odd, Sam will buy x chocolates; on days when x is even, Sam will not purchase any chocolates.

Complete the code in the editor so that for each day N_i (where $1 \leq x \leq N \leq Y$) in array `arr`, the number of chocolates Sam purchased (during days 1 through N) is printed on a new line. This is a function-only challenge, so input is handled for you by the locked stub code in the editor.

Input Format

The program takes an array of integers as a parameter.

The locked code in the editor handles reading the following input from `stdin`, assembling it into an array of integers (`arr`), and calling `calculate(arr)`.

The first line of input contains an integer, T (the number of test cases). Each line i of the T subsequent lines describes the i th test case as an integer, N_i (the number of days).

Constraints

$$1 \leq T \leq 2 \times 10^5$$

$$1 \leq N \leq 2 \times 10^6$$

$$1 \leq x \leq N \leq Y$$

Output Format

For each test case, T_i in `arr`, your `calculate` method should print the total number of chocolates Sam purchased by day N_i on a new line.

Sample Input 0

```
3
1
2
3
```

Sample Output 0

```
1
1
4
```

Program:

```
1  #include<stdio.h>
2  int main()
3  {
4      int t;
5      scanf("%d",&t);
6      while(t-->0)
7      {
8          int n,c=0;
9          scanf("%d",&n);
10         for(int i=0;i<=n;i++)
11         {
12             if(i%2!=0)
13                 c=c+i;
14         }
15         printf("%d\n",c);
16     }
17 }
```

	Input	Expected	Got	
✓	3	1	1	✓
	1	1	1	
	2	4	4	
	3			
✓	10	1296	1296	✓
	71	2500	2500	
	100	1849	1849	
	86	729	729	
	54	400	400	
	40	25	25	
	9	1521	1521	
	77	25	25	
	9	49	49	
	13	2401	2401	
	98			

Passed all tests! ✓

Ex.No. 36

Date 15.11.24

Football Scores

Problem Statement:

The number of goals achieved by two football teams in matches in a league is given in the form of two lists. Consider:

Football team A, has played three matches, and has scored { 1 , 2 , 3 } goals in each match respectively.

Football team B, has played two matches, and has scored { 2, 4 } goals in each match respectively.

Your task is to compute, for each match of team B, the total number of matches of team A, where team A has scored less than or equal to the number of goals scored by team B in that match.

In the above case:

For 2 goals scored by team B in its first match, team A has 2 matches with scores 1 and 2.

For 4 goals scored by team B in its second match, team A has 3 matches with scores 1, 2 and 3. Hence, the answer: {2, 3}.

Complete the code in the editor below. The program must return an array of m positive integers, one for each `maxes[i]` representing the total number of elements `nums[j]` satisfying `nums[j] ≤ maxes[i]` where $0 \leq j < n$ and $0 \leq i < m$, in the given order.

It has the following:

`nums[nums[0],...nums[n-1]]`: first array of positive integers

`maxes[maxes[0],...maxes[n-1]]`: second array of positive integers

Constraints:

$2 \leq n, m \leq 105$, $1 \leq \text{nums}[j] \leq 109$, where $0 \leq j < n$, $1 \leq \text{maxes}[i] \leq 109$, where $0 \leq i < m$.

Input Format For Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer n, the number of elements in nums.

The next n lines each contain an integer describing `nums[j]` where $0 \leq j < n$.

The next line contains an integer m, the number of elements in maxes.

The next m lines each contain an integer describing `maxes[i]` where $0 \leq i < m$.

Sample Input

```
4
1
4
2
4
2
3
5
```

Sample Output

```
2
4
```

Program:

```
1  #include<stdio.h>
2  int main()
3  {
4      int s1,s2,ans;
5      scanf("%d",&s1);
6      int ta[s1];
7      for(int i=0;i<s1;i++)
8          scanf("%d",&ta[i]);
9      scanf("%d",&s2);
10     int tb[s2];
11     for(int i=0;i<s2;i++)
12         scanf("%d",&tb[i]);
13     for(int j=0;j<s2;j++)
14     {
15         ans=0;
16         for(int i=0;i<s1;i++)
17         {
18             if(tb[j]>=ta[i])
19                 ans++;
20         }
21         printf("%d\n",ans);
22     }
23 }
```

	Input	Expected	Got	
✓	4	2	2	✓
	1	4	4	
	4			
	2			
	4			
	2			
	3			
	5			
✓	5	1	1	✓
	2	0	0	
	10	3	3	
	5	4	4	
	4			
	8			
	4			
	3			
	1			
	7			
	8			

Passed all tests! ✓